Ex 6a: Exploratory Data Analysis – Netflix Titles

Question

(An Autonomous Institution, Affiliated to Anna University, Chennai)
Department of Artificial Intelligence and Data Science
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AD23532 – Principles of Data Science

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Ex6a: Exploratory Data Analysis

Learning Objective:
The objective of this experiment is to enable students to understand and perform Exploratory Data Analysis (EDA)on a real-world Netflix titles dataset using Python.

Problem Statement:
Netflix maintains a large collection of movies and TV shows containing information such as title, content type, release year, date added, country, rating, duration, genre, director, cast, and description. Netflix wants to understand the characteristics and growth of its content library and identify important patterns in its movies and TV shows.

As a Data Scientist, perform an Exploratory Data Analysis on the Netflix Titles dataset. Analyse the content distribution, ratings, release and addition trends, duration, countries, genres, directors, actors, and textual descriptions. Use suitable statistical techniques and visualisations to identify patterns and generate meaningful insights from the data.

Tasks to Perform:

1. Download Netflix_titles dataset from kaggle and convert it to dataframe.
2. Explore the Dataset
3. **Content Distribution Analysis. (Plot graphs wherever needed)**
    a. How many Movies vs TV Shows are in the dataset?
    b. What is the distribution of ratings (e.g., TV-MA, PG, R, etc.)?
    c. Which countries have the most Netflix content?
4. **Time-Based Trend Analysis (Plot graphs wherever needed)**
    a. In which years were the most shows/movies released?
    b. In which years were titles added to Netflix?
    c. How has Netflix's content growth trend changed over time?
5. **Duration & Content Type Analysis (Plot graphs wherever needed)**
    a. What is the distribution of movie durations (in minutes)?
    b. How many TV shows exist per number of seasons?
6. **Genre & Categories Analysis (Plot graphs wherever needed)**
    a. What are the top 10 most common genres on Netflix?
    b. Which directors have the most titles on Netflix?
    c. Which actors appear most frequently across Netflix shows/movies?
7. **Text & Descriptions**
    a. What are the most common words in the description field? (word cloud)
8. In which months of the year does Netflix add the most content?
9. Which countries produce more movies vs TV shows**?**
10. What are the most common ratings for Movies vs TV Shows?
11. What are the oldest and newest titles available on Netflix?
12. Which genres frequently co-occur together**?** (Genre correlation heatmap)

Aim

To perform exploratory data analysis on the Netflix Titles dataset and identify patterns in content type, ratings, countries, release and addition trends, duration, genres, people and descriptions.

Procedure

1. Download netflix_titles.csv from Kaggle and load it into a DataFrame.
2. Explore the dataset: shape, info, missing values and summary statistics.
3. Analyse content distribution: Movies vs TV Shows, rating distribution and top countries.
4. Analyse time trends: release years, years added to Netflix and growth over time.
5. Analyse duration: movie runtime distribution and number of TV shows per season count.
6. Analyse genres, directors and actors (top 10 each) and build a word cloud of descriptions.
7. Find the best months for content additions, country-wise movies vs TV shows, ratings by type, oldest and newest titles, and a genre co-occurrence heatmap.

Code

```
#Mohammed Fayyad(241801165)
# !pip install wordcloud
import pandas as pd, numpy as np, matplotlib.pyplot as plt, seaborn as sns
from collections import Counter
from wordcloud import WordCloud

df = pd.read_csv("netflix_titles.csv")
print(df.shape); df.info(); print(df.isnull().sum());
print(df.describe(include="all"))

df["date_added"] = pd.to_datetime(df["date_added"].str.strip(), errors="coerce")
df["year_added"] = df["date_added"].dt.year
df["month_added"] = df["date_added"].dt.month_name()

# 3. Content distribution
print(df["type"].value_counts()); df["type"].value_counts().plot(kind="bar");
plt.show()
sns.countplot(y="rating", data=df, order=df["rating"].value_counts().index);
plt.show()
df["country"].dropna().str.split(",
").explode().value_counts().head(10).plot(kind="barh"); plt.show()

# 4. Time trends
df["release_year"].value_counts().sort_index().plot(); plt.title("Releases per
year"); plt.show()
df["year_added"].value_counts().sort_index().plot(kind="bar"); plt.title("Added per
year"); plt.show()
df.groupby(["year_added", "type"]).size().unstack().plot(); plt.title("Growth
trend"); plt.show()
```

```
# 5. Duration
movies = df[df.type == "Movie"].copy()
movies["mins"] = movies["duration"].str.replace(" min", "").astype(float)
sns.histplot(movies["mins"].dropna(), bins=30, kde=True); plt.show()
shows = df[df.type == "TV Show"]
shows["duration"].value_counts().plot(kind="bar"); plt.show()

# 6. Genres, directors, actors
genres = df["listed_in"].dropna().str.split(", ").explode()
genres.value_counts().head(10).plot(kind="barh"); plt.show()
df["director"].dropna().str.split(",
").explode().value_counts().head(10).plot(kind="barh"); plt.show()
df["cast"].dropna().str.split(",
").explode().value_counts().head(10).plot(kind="barh"); plt.show()

# 7. Word cloud
wc = WordCloud(width=900, height=450, background_color="white").generate("
".join(df["description"].dropna()))
plt.imshow(wc); plt.axis("off"); plt.show()

# 8. Months
df["month_added"].value_counts().plot(kind="bar"); plt.show()

# 9. Country vs type
ct = df.dropna(subset=["country"]).assign(country=lambda d:
d["country"].str.split(", ")).explode("country")
top = ct["country"].value_counts().head(10).index
pd.crosstab(ct[ct.country.isin(top)]["country"], ct["type"]).plot(kind="bar");
plt.show()

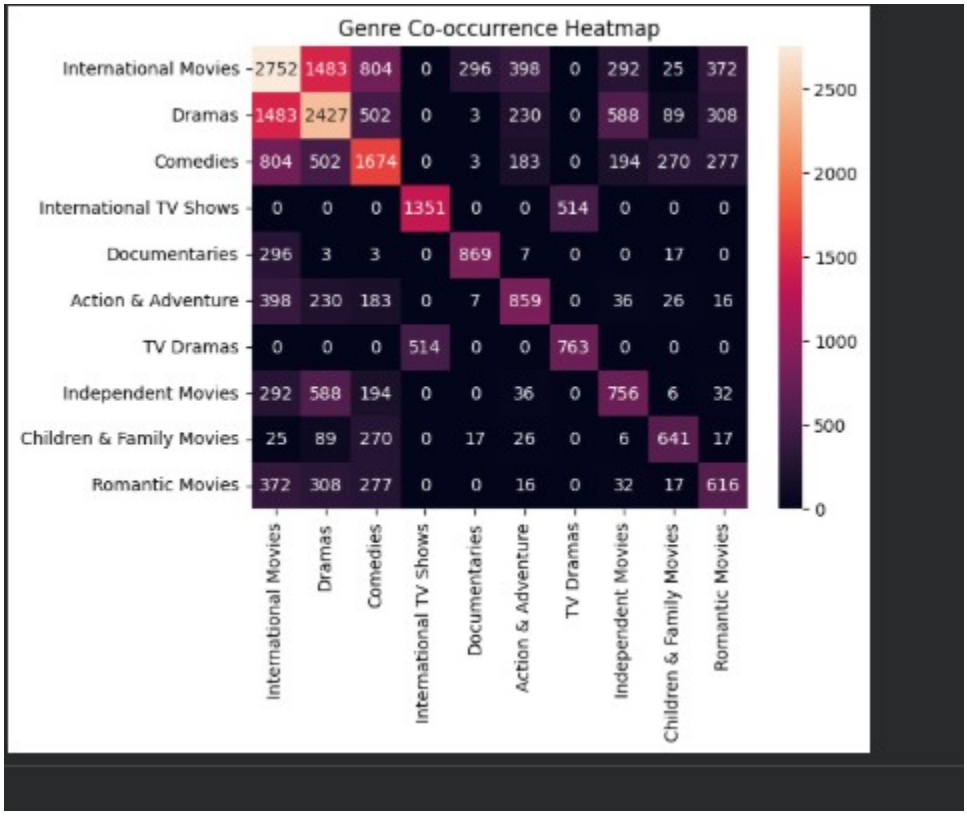
# 10. Ratings by type
pd.crosstab(df["rating"], df["type"]).plot(kind="bar", figsize=(10, 5)); plt.show()

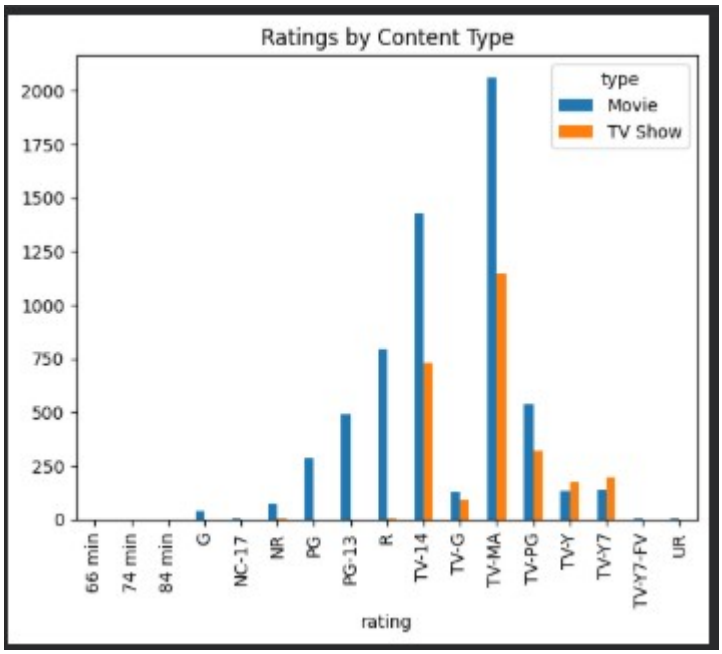
# 11. Oldest / newest
print(df.sort_values("release_year").head(5)[["title", "release_year"]])
print(df.sort_values("release_year", ascending=False).head(5)[["title",
"release_year"]])

# 12. Genre co-occurrence heatmap
g = df["listed_in"].dropna().str.split(", ")
top_g = genres.value_counts().head(15).index
mat = pd.DataFrame({x: g.apply(lambda l: int(x in l)) for x in top_g})
sns.heatmap(mat.corr(), cmap="coolwarm"); plt.show()
```

Result

EDA on the Netflix dataset was completed. The plots show the split between Movies and TV Shows, rating and country distributions, release and addition trends, duration patterns, the most common genres, directors and actors, the description word cloud, monthly addition pattern, oldest/newest titles and genre co-occurrence.


Genre Co-occurrence Heatmap
International Movies 2752 1483 804 0 296 398 0 292 25 372
Dramas 1483 2427 502 0 3 230 0 588 89 308
Comedies 804 502 1674 0 3 183 0 194 270 277
International TV Shows 0 0 0 1351 0 0 514 0 0 0
Documentaries 296 3 3 0 869 7 0 0 17 0
Action & Adventure 398 230 183 0 7 859 0 36 26 16
TV Dramas 0 0 0 514 0 0 763 0 0 0
Independent Movies 292 588 194 0 0 36 0 756 6 32
Children & Family Movies 25 89 270 0 17 26 0 6 641 17
Romantic Movies 372 308 277 0 0 16 0 32 17 616
International Movies
Dramas
Comedies
International TV Shows
Documentaries
Action & Adventure
TV Dramas
Independent Movies
Children & Family Movies
Romantic Movies
2500
2000
1500
1000
500
0


Ratings by Content Type
type
Movie
TV Show
2000
1750
1500
1250
1000
750
500
250
0
66 min
74 min
84 min
G
NC-17
NR
PG
PG-13
R
TV-14
TV-G
TV-MA
TV-PG
TV-Y
TV-Y7
TV-Y7-FV
UR
rating